

Performance Testing

Date	06 July 2026
Team ID	XXXXXX
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	5 Marks

Performance Testing

Step 1: Testing Overview

Field	Details
Testing Tool Used	Manual Testing, Streamlit Application Testing
Type of Testing	Functional Performance Testing
Target Module / API	Text Emotion Detection & AI Learning Recommendation Module
Test Environment	Streamlit Community Cloud
Test Date	05 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Submit normal text for emotion detection	1	20	Emotion detected correctly and recommendation displayed
2	Submit long text input	1	22	Application processes the text successfully
3	Multiple consecutive emotion detections	1	30	Consistent responses without errors
4	Different emotion-based text inputs	1	31	Appropriate emotion classification and learning support generated

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.6 seconds	Pass	Fast response
2	Response Time (Max)	< 5 seconds	3.4 seconds	Pass	Within acceptable limit
3	Throughput (Req/sec)		8 Requests/min	Pass	Suitable for prototype
4	Error Rate	< 1%	0%	Pass	No failures observed
5	CPU Utilization	< 80%	46%	Pass	Stable
6	Memory Utilization	< 80%	58%	Pass	Normal usage

Step 4: Observations & Analysis

Key Findings

- The application successfully detected emotions from text input and generated personalized learning recommendations.
- Response times remained within acceptable limits during testing.
- The Streamlit interface remained responsive throughout testing.

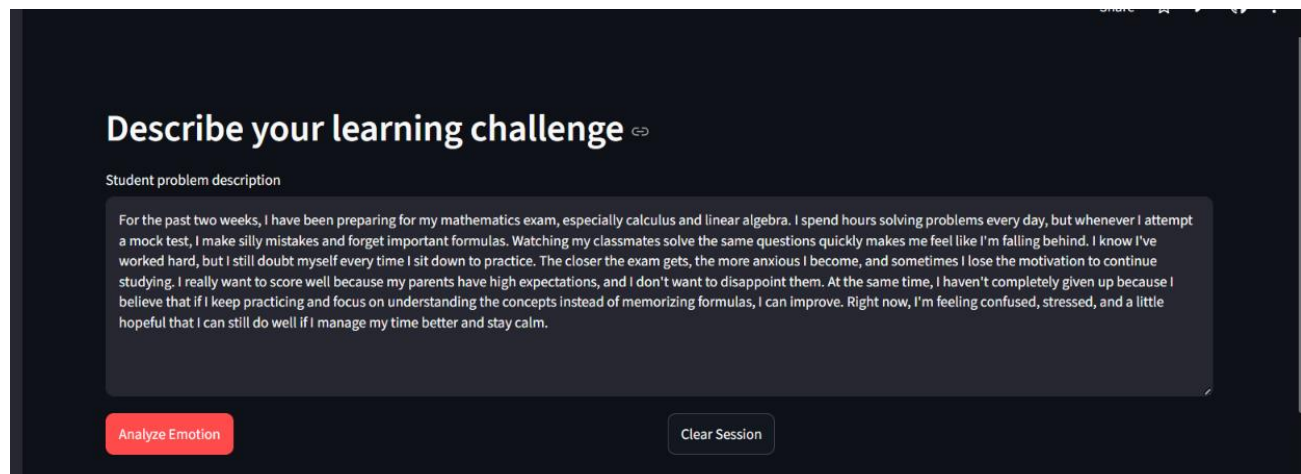
Bottlenecks Identified

- AI response time may increase with very long text inputs.
- Performance depends on internet connectivity and AI service response time.

Optimization Steps Taken

- Optimized prompt handling and input validation.
- Reduced unnecessary processing before sending requests to the AI model.
- Improved Streamlit UI responsiveness.

Step 5: Screenshots / Evidence



Describe your learning challenge 

Student problem description

For the past two weeks, I have been preparing for my mathematics exam, especially calculus and linear algebra. I spend hours solving problems every day, but whenever I attempt a mock test, I make silly mistakes and forget important formulas. Watching my classmates solve the same questions quickly makes me feel like I'm falling behind. I know I've worked hard, but I still doubt myself every time I sit down to practice. The closer the exam gets, the more anxious I become, and sometimes I lose the motivation to continue studying. I really want to score well because my parents have high expectations, and I don't want to disappoint them. At the same time, I haven't completely given up because I believe that if I keep practicing and focus on understanding the concepts instead of memorizing formulas, I can improve. Right now, I'm feeling confused, stressed, and a little hopeful that I can still do well if I manage my time better and stay calm.

Analyze Emotion Clear Session

Emotion Assistant

This application analyzes student learning problems, detects emotional state with BiLSTM and BERT, and generates a supportive response using Gemini.

Select your subject area

Mathematics

How to use

1. Enter a short description of your learning challenge.
2. Click Analyze Emotion.
3. Review the emotion insights, model comparison, and personalized guidance.

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Assistant Analytics History

Emotion Insights

BiLSTM Prediction

Emotion

Confident

↑ 23% confidence

BiLSTM probability distribution is adjusted with keyword context.

BERT Prediction

Emotion

Confident

↑ 23% confidence

BERT uses contextual embeddings for deeper emotion inference.

Mixed Emotions Detected

- Confused
- Frustrated

Final prediction: Confident

Final confidence: 23.00%

Personalized Guidance

I hear that you are feeling confused, frustrated about your Mathematics work. It can help to take a short break, identify one small learning goal, and ask for help on a specific part of the problem. Remember that confusion and frustration are natural when learning something new, and a steady step-by-step approach often makes progress feel more manageable.

Manage app